

Assignment-19.1

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Task 1 – Python to Java Conversion

Task:

Translate a simple Python class into Java using AI assistance.

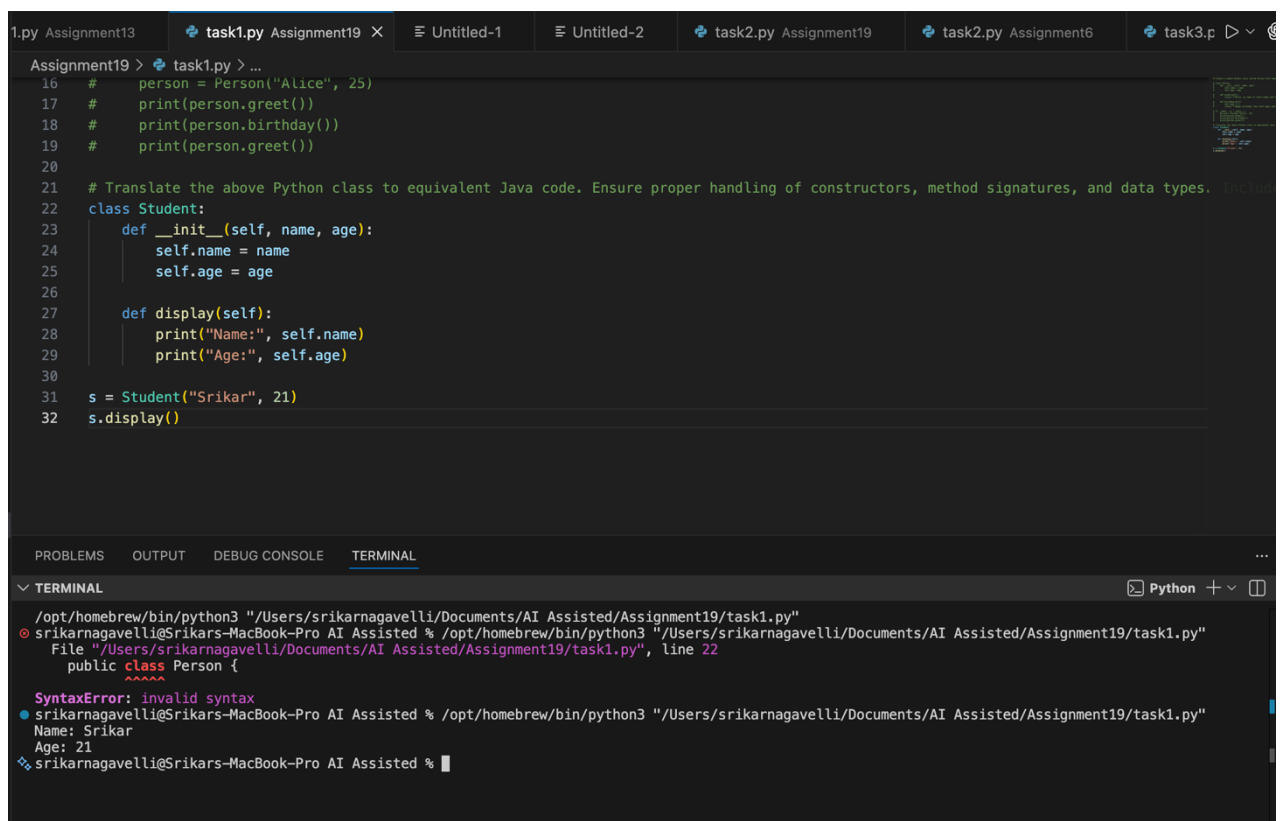
Instructions:

- Prompt AI to convert a Python class with attributes and methods into equivalent Java code.
- Verify that constructors, method signatures, and data types are properly handled.
- Run both versions (Python & Java) to compare expected outputs.

Expected Output:

- Equivalent working Java class translated from Python.

CODE:



```
1.py Assignment13 task1.py Assignment19 X Untitled-1 Untitled-2 task2.py Assignment19 task2.py Assignment6 task3.p
Assignment19 > task1.py > ...
16 # person = Person("Alice", 25)
17 # print(person.greet())
18 # print(person.birthday())
19 # print(person.greet())
20
21 # Translate the above Python class to equivalent Java code. Ensure proper handling of constructors, method signatures, and data types. Include
22 class Student:
23     def __init__(self, name, age):
24         self.name = name
25         self.age = age
26
27     def display(self):
28         print("Name:", self.name)
29         print("Age:", self.age)
30
31 s = Student("Srikar", 21)
32 s.display()
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
▼ TERMINAL Python + ▾
/opt/homebrew/bin/python3 "/Users/srikarnagavelli/Documents/AI Assisted/Assignment19/task1.py"
srikarnagavelli@Srikars-MacBook-Pro AI Assisted % /opt/homebrew/bin/python3 "/Users/srikarnagavelli/Documents/AI Assisted/Assignment19/task1.py"
File "/Users/srikarnagavelli/Documents/AI Assisted/Assignment19/task1.py", line 22
    public class Person {
           ^^^^^
SyntaxError: invalid syntax
srikarnagavelli@Srikars-MacBook-Pro AI Assisted % /opt/homebrew/bin/python3 "/Users/srikarnagavelli/Documents/AI Assisted/Assignment19/task1.py"
Name: Srikar
Age: 21
srikarnagavelli@Srikars-MacBook-Pro AI Assisted %
```

```
Assignment19 > J Person.java
1  class Student {
2      String name;
3      int age;
4
5      Student(String name, int age) {
6          this.name = name;
7          this.age = age;
8      }
9
10     // Method
11     void display() {
12         System.out.println("Name: " + name);
13         System.out.println("Age: " + age);
14     }
15
16     public static void main(String[] args) {
17         Student s = new Student("Srikar", 21);
18         s.display();
19     }
20 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

TERMINAL

```
/opt/homebrew/bin/python3 "/Users/srikarnagavelli/Documents/AI Assisted/Assignment19/tas
srikarnagavelli@Srikars-MacBook-Pro AI Assisted % /opt/homebrew/bin/python3 "/Users/srik
File "/Users/srikarnagavelli/Documents/AI Assisted/Assignment19/task1.py", line 22
    public class Person {
        ^^^^^
SyntaxError: invalid syntax
srikarnagavelli@Srikars-MacBook-Pro AI Assisted % /opt/homebrew/bin/python3 "/Users/srik
Name: Srikar
Age: 21
srikarnagavelli@Srikars-MacBook-Pro AI Assisted % /opt/homebrew/bin/python3 "/Users/srik
Name: Srikar
Age: 21
srikarnagavelli@Srikars-MacBook-Pro AI Assisted %
```

Task 2 – C++ to Python Function Conversion

Task:

Convert a C++ function for factorial calculation into Python.

Instructions:

- Ask AI to translate a recursive C++ factorial function into Python.
- Validate correctness by testing with sample inputs.
- Ensure Python version follows Pythonic conventions.

Expected Output:

- A correct Python implementation equivalent to the original C++ factorial function.

CODE:

task1.py Assignment19 Person.java Untitled-1 Untitled-2 task2.py Assignment1

Assignment19 > task2.py > ...
1 def factorial(n):
2 if n == 0:
3 return 1
4 else:
5 return n * factorial(n - 1)
6 number = 5
7 print(f"The factorial of {number} is {factorial(number)}")

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

▼ TERMINAL
/opt/homebrew/bin/python3 "/Users/srikarnagavelli/Documents/AI Assisted/Assignment19/task1.py"
srikarnagavelli@Srikars-MacBook-Pro AI Assisted % /opt/homebrew/bin/python3 "/Users/srikarnagavelli/D
File "/Users/srikarnagavelli/Documents/AI Assisted/Assignment19/task1.py", line 22
 public class Person {
 ^^^^^
SyntaxError: invalid syntax
srikarnagavelli@Srikars-MacBook-Pro AI Assisted % /opt/homebrew/bin/python3 "/Users/srikarnagavelli/D
Name: Srikar
Age: 21
srikarnagavelli@Srikars-MacBook-Pro AI Assisted % /opt/homebrew/bin/python3 "/Users/srikarnagavelli/D
Name: Srikar
Age: 21
srikarnagavelli@Srikars-MacBook-Pro AI Assisted % /opt/homebrew/bin/python3 "/Users/srikarnagavelli/D
The factorial of 5 is 120
srikarnagavelli@Srikars-MacBook-Pro AI Assisted %

Task

```
task_1.py task_1.java task_2.c++ task2.py launch.json
task_2.c++ > main()
1 #include <iostream>
2 using namespace std;
3
4 int factorial(int n) {
5     if(n == 0)
6         return 1;
7     else
8         return n * factorial(n-1);
9 }
10
11 int main() {
12     cout << factorial(5);
13     return 0;
14 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

> TERMINAL

```
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python39\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task_1.py
Name: Akram
Age: 21
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python39\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task2.py
120
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python39\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task2.py
120
PS C:\Users\akram\OneDrive\Desktop\AIAC> 
```

Connect Indexing completed. Java: Ready Ln 14, Col 2 Sp

3 – SQL to Pandas Query

Task:

Translate a SQL query into a Pandas DataFrame operation in Python.

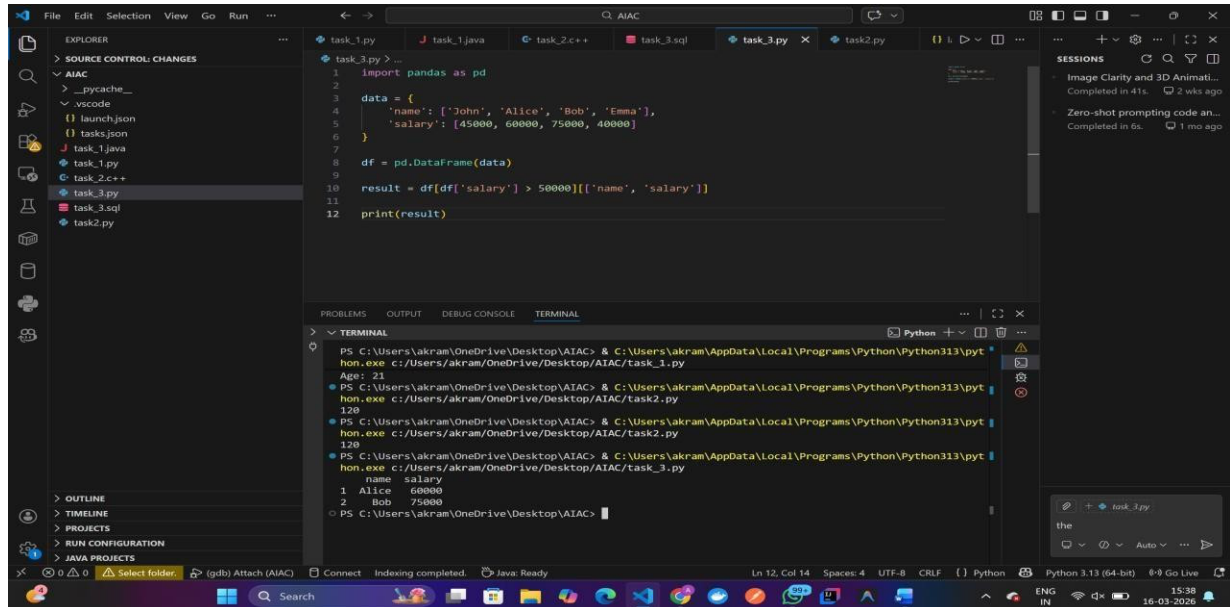
Instructions:

- Provide AI with a SQL query (e.g., SELECT name, salary FROM employees WHERE salary > 50000).
- Ask AI to generate equivalent Pandas code.
- Test the generated code on a sample DataFrame.

Expected Output:

- Equivalent Pandas query that retrieves the correct subset of data.

CODE:

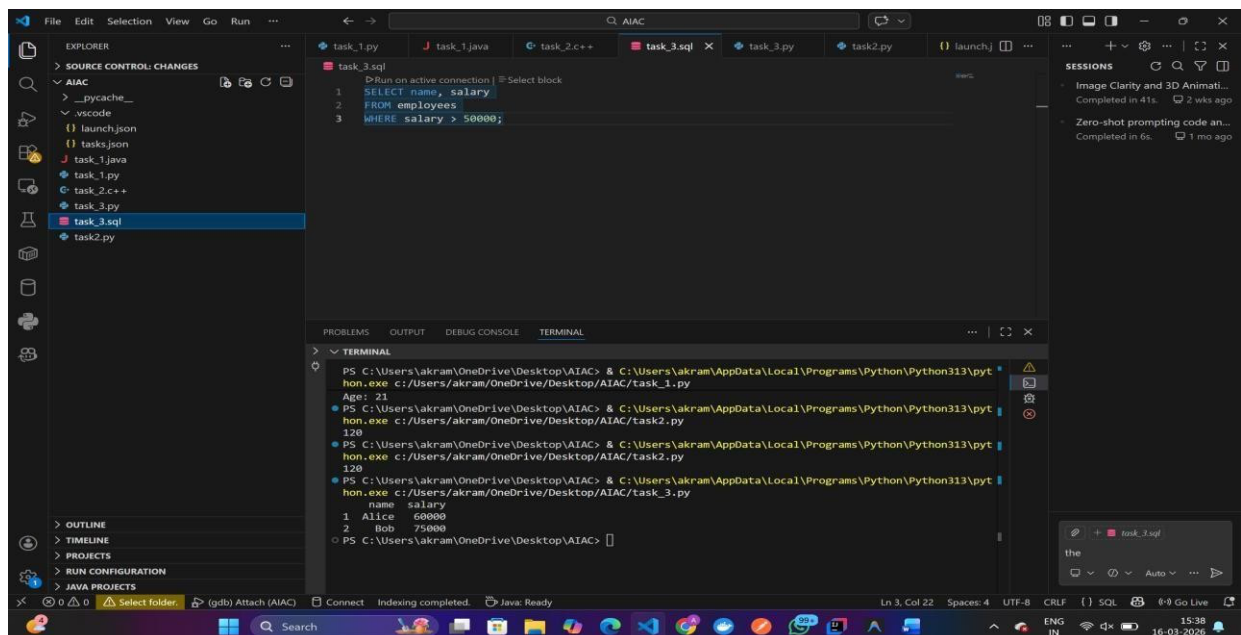


The screenshot shows the VS Code editor with a file explorer on the left displaying a project named 'AIAC'. The main editor window shows a Python file named 'task_3.py' with the following code:

```
1 import pandas as pd
2
3 data = {
4     'name': ['John', 'Alice', 'Bob', 'Emma'],
5     'salary': [45000, 60000, 75000, 40000]
6 }
7
8 df = pd.DataFrame(data)
9
10 result = df[df['salary'] > 50000][['name', 'salary']]
11
12 print(result)
```

The terminal at the bottom shows the execution of the script, displaying the output:

```
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task_3.py
Age: 21
name salary
1 Alice 60000
2 Bob 75000
```



The screenshot shows the VS Code editor with a file explorer on the left displaying a project named 'AIAC'. The main editor window shows a SQL file named 'task_3.sql' with the following code:

```
1 SELECT name, salary
2 FROM employees
3 WHERE salary > 50000;
```

The terminal at the bottom shows the execution of the script, displaying the output:

```
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task_3.py
Age: 21
name salary
1 Alice 60000
2 Bob 75000
```

Task 4 – Real-Time Application: Multi-Language Snippet

Converter

Scenario:

Students will choose a real-time use case where they need code in multiple languages. Example:

- Converting a Python web scraping snippet into JavaScript for browser automation.

- Translating a Java method into C# for enterprise applications.

Instructions:

- Prompt AI to translate code between two chosen languages.
- Test both implementations to ensure they produce the same results.
- Document challenges faced during translation (e.g., syntax, libraries).

Expected Output:

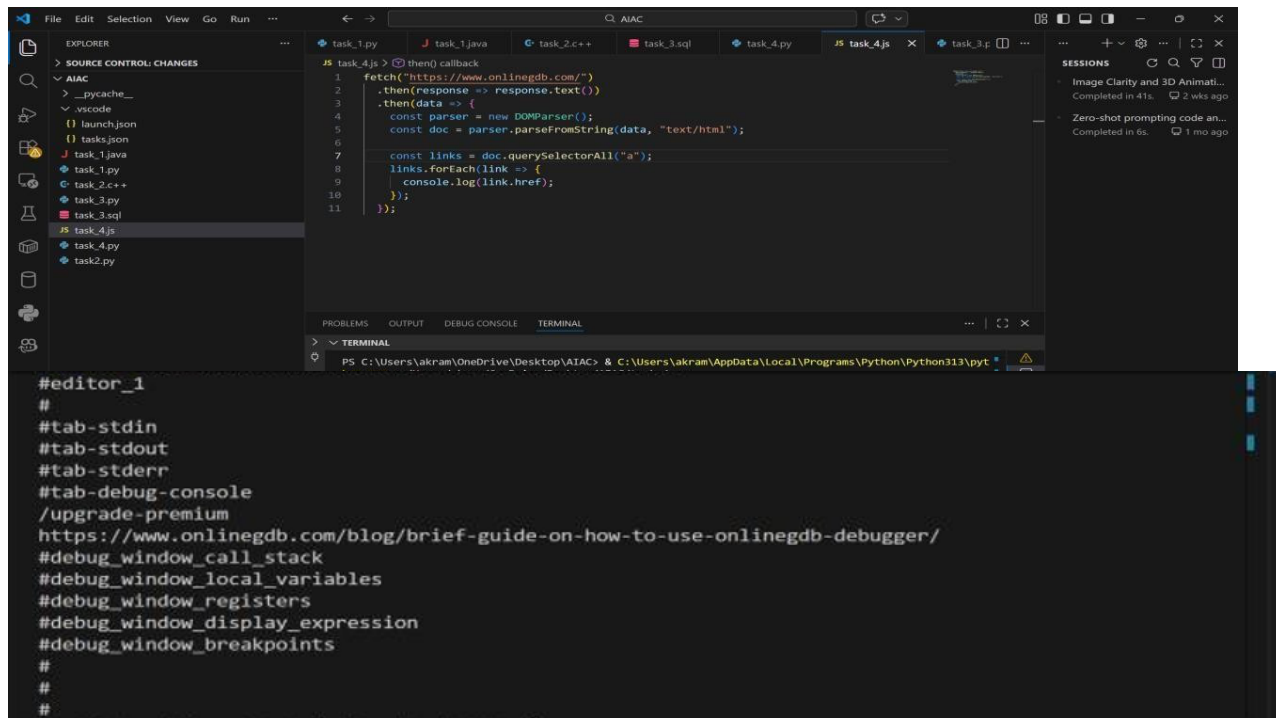
- Working equivalent code snippets in at least two different programming languages

CODE:

The screenshot shows a Visual Studio Code editor window with a Python file named `task_4.py` open. The code is a web scraping script using `requests` and `BeautifulSoup` to fetch a page and extract a link. The terminal at the bottom shows the command prompt and the output of the script, which is a URL from the onlinegdb website. The Explorer sidebar on the left shows a project structure with various files and folders. The Output window on the right shows the execution output of the script.

```
1 import requests
2 from bs4 import BeautifulSoup
3
4 url = "https://www.onlinegdb.com/"
5
6 response = requests.get(url)
7 soup = BeautifulSoup(response.text, "html.parser")
8
9 for link in soup.find_all('a'):
10     print(link.get('href'))
```

```
#editor_1
#
#tab-stdin
#tab-stdout
#tab-stderr
#tab-debug-console
/upgrade-premium
https://www.onlinegdb.com/blog/brief-guide-on-how-to-use-onlinegdb-debugger/
#debug_window_call_stack
#debug_window_local_variables
#debug_window_registers
#debug_window_display_expression
#debug_window_breakpoints
#
#
#
```



Explanation: In this assignment, we used AI to convert code from one programming language to another. We converted Python to Java, C++ to Python, SQL to Pandas, and also worked on a real-time example. We checked that all converted programs give correct output. This helped us understand differences between languages and how AI can make coding easier and faster.